

BIOS 755: Linear Mixed Models I

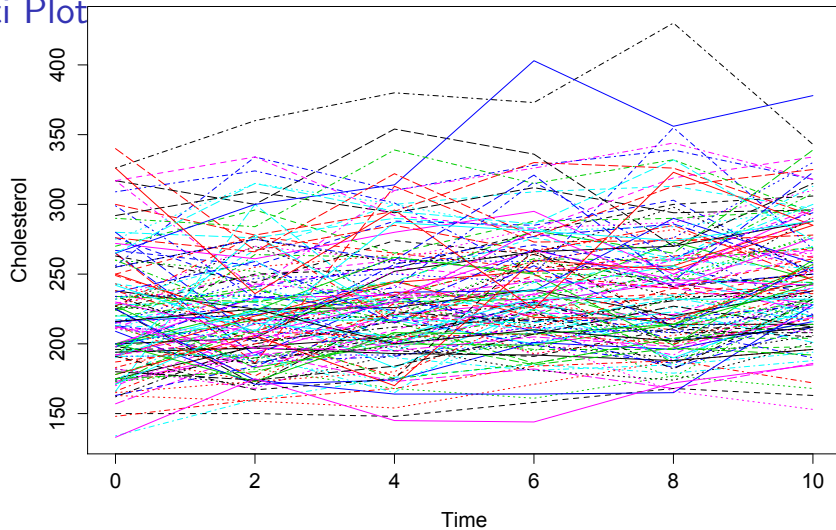
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Framingham study Cholesterol Data

- ▶ In the Framingham study, each of 2634 participants was examined every 2 years for a 10 year period for his/her cholesterol level.
- ▶ Study objectives:
 - ▶ How does cholesterol level change over time on average as people get older?
 - ▶ How is the change of cholesterol level associated with sex and baseline age?
- ▶ A subset of 200 subjects' data is used for illustrative purpose.

Spaghetti Plot



Introduction to Linear Mixed Models

- ▶ In the General Linear Model, we focused our conceptual model on the covariance and correlation of the error terms.
- ▶ In linear mixed models, the conceptual model is based on thinking about individual behavior first.
- ▶ The possibilities for how this is represented and how the variation in the population is represented are very flexible.
- ▶ As we'll see, linear mixed models can incorporate heterogeneity and different correlation structures (even though we don't think about them that way).

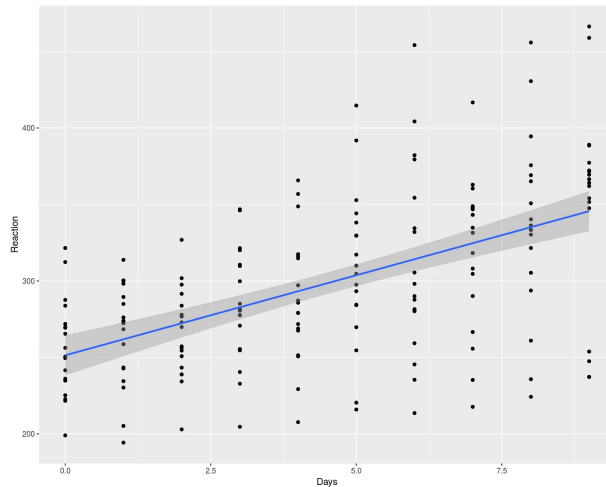
Linear Mixed (Effects) Models

- ▶ The linear mixed model can be expressed as

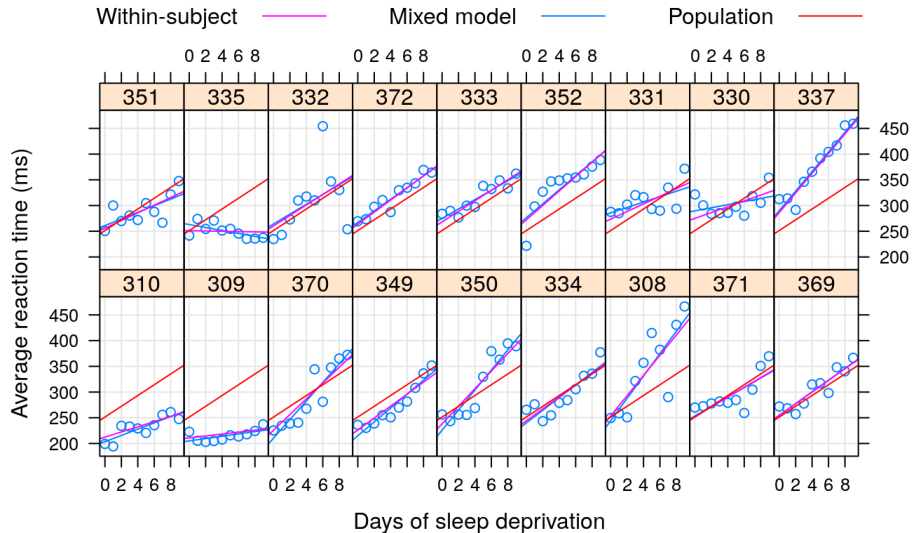
$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \mathbf{e}_i$$

where

- ▶ $\mathbf{X}_i$ – $n_i \times p$ matrix of fixed effect covariates
- ▶ $\boldsymbol{\beta}$ – $k \times 1$ vector of regression coefficients (fixed effects).
- ▶ $\mathbf{Z}_i$ – $n_i \times q$ matrix of random effect covariates.
- ▶ $\mathbf{b}_i$ – $q \times 1$ vector of random effects, $\mathbf{b}_i \sim N(0, \mathbf{G})$,
- ▶ $\mathbf{e}_i$ – $n_i \times 1$ vector of errors and $\mathbf{e}_i \sim N(0, \mathbf{R}_i)$.



- Consider a sleep deprivation study where the sleeping time of 18 individuals was restricted, and their reaction to a series of tests was measured over 10 days.



Random intercept and slope model

The random intercept and slope model:

$$\mathbf{Y}_i = \beta_0 + b_{0i} + (\beta_1 + b_{1i})\mathbf{t}_i + \mathbf{e}_i$$

where $\mathbf{t}'_i = \{t_{i1}, t_{i2}, \dots, t_{in_i}\}$

- ▶ β_0 is the average intercept and b_{0i} are the deviations from the average intercept.
- ▶ β_1 is the average slope and b_{1i} are the deviations from the average slope.
- ▶ We could add other fixed effects to this model (sex, smoking, etc.).

Random intercept and slope model

The random intercept and slope model:

$$\mathbf{Y}_i = \beta_0 + b_{0i} + (\beta_1 + b_{1i})\mathbf{t}_i + \mathbf{e}_i$$

- ▶ $\mathbf{R}_i = \text{var}(\mathbf{e}_i)$ describes the covariance of the residuals
- ▶ In the models we've been running in the previous weeks, this is the covariance of the i th subject's deviations from $\beta_0 + \beta_1\mathbf{t}_i$ (i.e., the overall trend)
- ▶ Now it's the covariance of the i th subject's deviations from $\beta_0 + b_{0i} + (\beta_1 + b_{1i})\mathbf{t}_i$ (i.e., their individual trend)
 - ▶ Usually, it is assumed that $\mathbf{R}_i = \sigma^2\mathbf{I}$, which is the “conditional independence assumption.”

Linear Mixed (Effects) Models

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \mathbf{e}_i$$

The vector of regression parameters $\boldsymbol{\beta}$ are the fixed effects, which are assumed to be the same for all individuals.

- ▶ Fixed effects are constant across individuals, and random effects vary.
- ▶ For example, in a growth study, a model with random intercepts $\beta_0 + b_{0i}$ and fixed slope β_1 corresponds to parallel lines for different individuals i , or the model $Y_{ij} = \beta_0 + b_{0i} + \beta_1 t_{ij} + e_{ij}$

Decomposing the Variation



- In the linear mixed-effects model

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \mathbf{e}_i,$$

the error term e_{ij} is decomposed as

$$e_{ij} = e_{ij1} + e_{ij2}$$

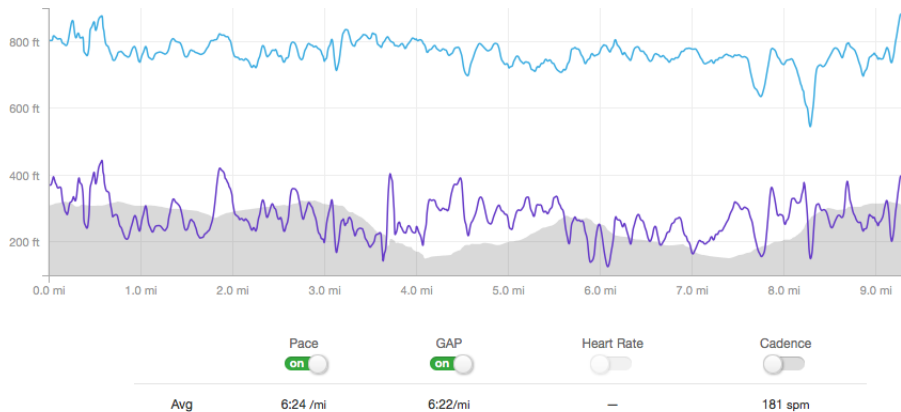
where e_{ij1} represents the deviations due to within-subject fluctuations and e_{ij2} those due to measurement error, where

$$\mathbf{e}_i = \mathbf{e}_{i1} + \mathbf{e}_{i2}.$$

Decomposing Variation



Decomposing Variation



Within-unit Variation

- ▶ Some typical scenarios; considerations involved in identifying an appropriate $\mathbf{R}_i$.
- ▶ There may be biological fluctuations over time, but commonly, the observation times are not close enough to catch these variations.
- ▶ Then correlation due to within-subject sources among the Y_{ij} may be considered negligible.
- ▶ If we furthermore believe that the magnitude of fluctuations is similar across time and units, we may represent this variance as

$$\text{var}(\mathbf{e}_{i1}) = \sigma_1^2 \mathbf{I}$$

Within-unit Variation

- ▶ It's probably reasonable to assume that errors in measurement are uncorrelated over time. Thus,

$$\text{var}(\mathbf{e}_{i2}) = \sigma_2^2 \mathbf{I}.$$

- ▶ We then have

$$\mathbf{R}_i = \text{var}(\mathbf{e}_i) = \text{var}(e_{1i}) + \text{var}(e_{2i}) = \sigma_1^2 \mathbf{I}_{n_i} + \sigma_2^2 \mathbf{I}_{n_i} = \sigma^2 \mathbf{I}_{n_i},$$

where σ^2 is the aggregate variance reflecting variation due to both within-unit sources.

- ▶ The assumption that e_{ij2} and e_{ij2} are independent is standard, as is the assumption that $\mathbf{e}_{i1}$ and $\mathbf{e}_{i2}$ (and hence $\mathbf{e}_i$) are independent of $\mathbf{b}_i$.

Within-unit Variation

The two special cases of within-unit variation:

- ▶ If there is no (or very little) measurement error (e.g. height and weight), $\mathbf{e}_i = \mathbf{e}_{i1}$ (all within-unit variation is due to things like “fluctuations”).
- ▶ Similarly, we may have a rather “noisy” measuring device such that, relative to errors in measurement, deviations due to within-unit subjects are virtually negligible. In this case, $\mathbf{e}_i = \mathbf{e}_{i2}$ (all within-unit variation is solely the measurement error variance).

Among-unit Variation



- ▶ The random effects $\mathbf{b}_i$ have mean 0 and represent variation resulting from individual units' differences.
- ▶ Intercepts and slopes may tend to be large or small together, so subjects with steeper slopes tend to “start out” larger at the beginning.
- ▶ This suggests that it would not necessarily be smart to think of $\text{var}(\mathbf{b}_i)$ as a diagonal matrix (independence).

Among-unit Variation

- For this reason, we can also specify a covariance matrix for the random effects.

$$\text{var}(\mathbf{b}_i) = \mathbf{G}$$

- For $\mathbf{b}_i = \{b_{0i}, b_{1i}\}'$, $\mathbf{G}$ is a 2×2 matrix

$$\mathbf{G} = \begin{pmatrix} G_{11} & G_{12} \\ G_{11} & G_{22} \end{pmatrix}$$

with

$$\text{var}(b_{0i}) = G_{11}, \quad \text{var}(b_{1i}) = G_{22}$$

$$\text{cov}(b_{0i}, b_{1i}) = G_{12}$$

Among-unit Variation

- ▶ A standard assumption is that the $\mathbf{b}_i$ have a multivariate normal distribution

$$\mathbf{b}_i \sim MVN(\mathbf{0}, \mathbf{G})$$

- ▶ It is usually assumed that $\mathbf{e}_i$ and $\mathbf{b}_i$ are independent. This says that the magnitude of variation within a unit does not depend on the magnitude of $\mathbf{b}_i$ for that unit.

Conditional vs marginal mean

- ▶ The **conditional** mean of $\mathbf{Y}_i$, given $\mathbf{b}_i$, is

$$E(\mathbf{Y}_i | \mathbf{b}_i) = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \mathbf{b}_i$$

- ▶ The **marginal** or **population-averaged** mean of $\mathbf{Y}_i$ is

$$E(\mathbf{Y}_i) = \mathbf{X}_i \boldsymbol{\beta}$$

- ▶ In contrast to $\boldsymbol{\beta}$, the vector $\mathbf{b}_i$ is comprised of subject-specific regression coefficients.
- ▶ All covariates in $\mathbf{Z}$ will be in $\mathbf{X}$, and it's rare to consider more than 2 variables in $\mathbf{Z}$.

Conditional vs marginal variance



- ▶ In the mixed model

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \mathbf{e}_i$$

- ▶ We have the following conditional and marginal expectations

$$E(\mathbf{Y}_i|\mathbf{b}_i) = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i, \quad E(\mathbf{Y}_i) = \mathbf{X}_i\boldsymbol{\beta}$$

along with the following conditional and marginal variances

$$\begin{aligned} \text{var}(\mathbf{Y}_i|\mathbf{b}_i) &= \text{var}(\mathbf{e}_i) = \mathbf{R}_i, \quad \text{and} \\ \text{var}(\mathbf{Y}_i) &= \text{var}(\mathbf{Z}_i\mathbf{b}_i) + \text{var}(\mathbf{e}_i) \\ &= \mathbf{Z}_i\mathbf{G}\mathbf{Z}_i' + \mathbf{R}_i \end{aligned}$$

Linear Mixed (Effects) Models

- ▶ Introducing random effects induces correlation among the $\mathbf{Y}_i$.
- ▶ $\text{Var}(\mathbf{Y}_i)$ is described in terms of a set of covariance parameters, some defining $\mathbf{G}$ and some defining $\mathbf{R}_i$.
- ▶ It is difficult to disentangle the variance for $\mathbf{G}$ and variance for $\mathbf{R}_i$, which is one reason why we usually assume $\mathbf{R}_i = \sigma^2 \mathbf{I}$
- ▶ Linear mixed models are just another type of covariance matrix, which can lead to strange results (as we'll see).

Linear Mixed (Effects) Models Summary



- ▶ LMMs account for correlation through random effects that are unique to each individual.
- ▶ LMMs offer flexibility in modeling different data types and can handle unbalanced designs much better than covariance pattern models.
- ▶ The interpretation of the fixed effects is similar to that in standard linear regression.
- ▶ LMMs come with assumptions such as normality of residuals, independence of errors, and homoscedasticity (constant variance of errors), which must be checked for valid inferences.
- ▶ LMMs are especially useful for hierarchical or multilevel data.